Cover letter

Dear Editors,

Please consider “Limited-label cross-session motor-imagery EEG recalibration: spatial refitting and the added value of Euclidean alignment” as a Paper for Journal of Neural Engineering. The manuscript examines what a short calibration run contributes after a longitudinal motor-imagery decoder has already received the same source and current-session labels. It compares frozen CSP/LDA, classifier-only recalibration, full spatial-filter/classifier refitting and Euclidean alignment followed by full refitting.

EA added 5.30 balanced-accuracy points beyond full refitting in the original twelve-person Kumar cohort and 5.65 points in the nine-person BNCI cohort. Weak-class recall increased by 10.45 and 8.52 points, respectively. Matched-coordinate, RMS and diagonal-reference controls examine the contribution of the alignment operation; prediction transitions identify corrected and introduced errors. The paired Kumar window analysis also establishes a material signal-definition dependence: the added EA contrast was 7.85 points for cue-anchored five-second windows and 0.13 points for matched task-start two-second windows. Participant-level analyses retain these results alongside their uncertainty and longitudinal distribution.

The contribution is an information-matched analysis of recalibration operations, class coverage and signal definitions using established alignment and decoding methods. This combination connects neural signal processing with the practical allocation of short calibration recordings. We intend single-anonymous review and provide a complete manuscript PDF with embedded figures and tables, supplementary methods, and the scientific data/code archive.

A related manuscript, PONE-D-26-52050, remains under consideration at PLOS ONE. It addresses the contribution of spatial-filter refitting in Kumar2024. The present manuscript shares Kumar participants, partitions, baseline predictions and previously reported component contrasts, and adds the alignment-centered question and analyses summarized in the attached Related Manuscript Statement. That statement identifies the archived source available for comparison; it does not claim access to current files in the PLOS submission system or that the manuscripts have received an editorial overlap assessment. The actual related submission should accompany the final disclosure.

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Sincerely,

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